Leveraged — borrow low, lend high

Post equity → borrow cheap at the Low venue → supply it dear at the High venue → pocket the spread, **multiplied by leverage**. Then watch leverage cut the other way.

EQUITY (YOURS)	LEVERAGE	CARRY SPREAD	YOUR ROE	HEALTH FACTOR
0	3.0×	+14.00%	+47.00%	—

TIME
0 d

equity **10,000** → collateral → borrow **0** @ **5.00%** (Low) → supply → position **0** @ **19.00%** (High)

▶ **Run time** speed 20 d/s leverage 3.0×

Open carry ✓ **Close position** **Reset**

① Borrow here — Low venue

5.00% borrow APR

borrow rate (drag = market moves)

your debt D **0 tUSDC**

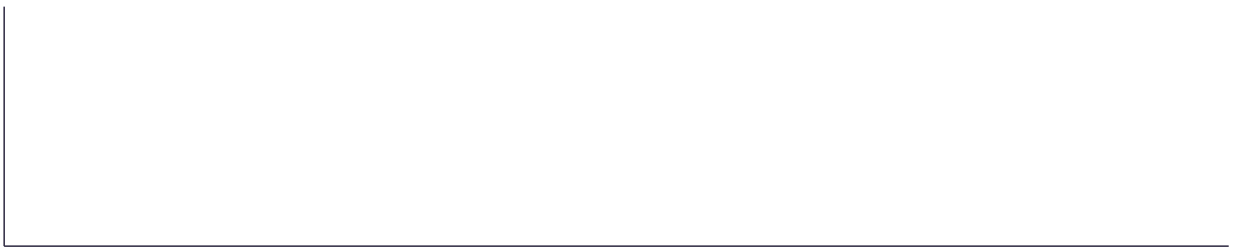
② Supply here — High venue

19.00% supply APR

supply rate (drag = market moves)

your position P **0 tUSDC**

Your equity (leveraged) Unleveraged (just supply equity to High)



Set a leverage and hit **Open carry**. At **3.0×** with spread **+14.00%**, your ROE would start at **+47.00%** — vs just **19.00%** unleveraged. That extra is the spread × (leverage−1).

The formula: ROE = s_{high} + (s_{high} − b_{low}) × (leverage − 1). At 1× you just earn the high supply rate; each extra turn adds one more **spread**. That is why degens loop.

Leverage has a ceiling: the position collateralizes the loan, so LTV = (ℓ−1)/ℓ must stay under the collateral factor → **ℓ ≤ 1/(1−CF)** (CF 0.85 → ~6×). At the ceiling HF≈1: one small adverse move and you are liquidated.

Leverage cuts both ways. Drag the **Low** borrow rate above the **High** supply rate (a rate shock): the spread goes negative, debt outgrows the position, HF slides to 1, and your equity is wiped **faster the more leverage you used**. Fragility = leverage (§28).

The dream → flash loan. This carry needs equity **locked as a cushion** and can be liquidated. What if you could borrow the whole position with **zero equity**, capture the spread, and repay in the **same transaction**? No cushion, no liquidation (atomic) — but only for instant, self-closing arbs. That is the flash loan, next.

MB/CSE 599 · AI Finance · leveraged rate carry (cross-venue) · leverage amplifies the spread — and the death